Supplementary Materials for: Cross-Modal Representational Alignment with LLM Priors for Image Generation

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A Architectures

A.1 Visualizations

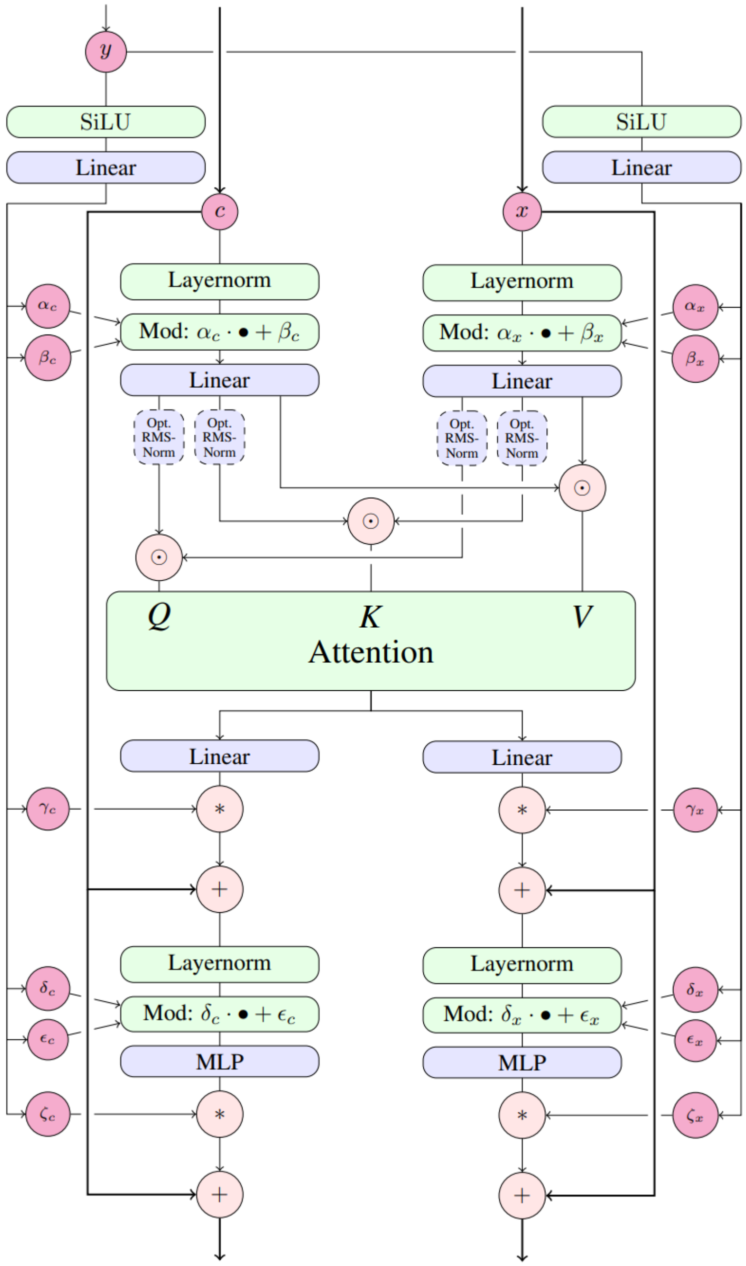

Figure 1: Dual-Stream Encoder architecture. Originally, MM-DiT block from Esser et al. [2024].

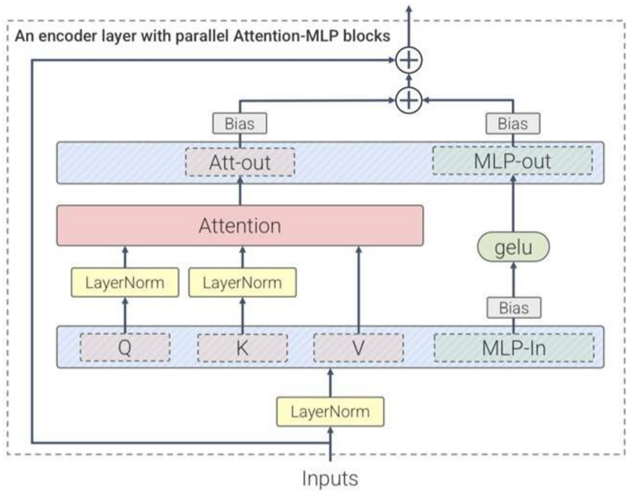

Figure 2: Single-Stream Decoder architecture. Originally, parallel ViT-22B layer Dehghani et al. [2023].

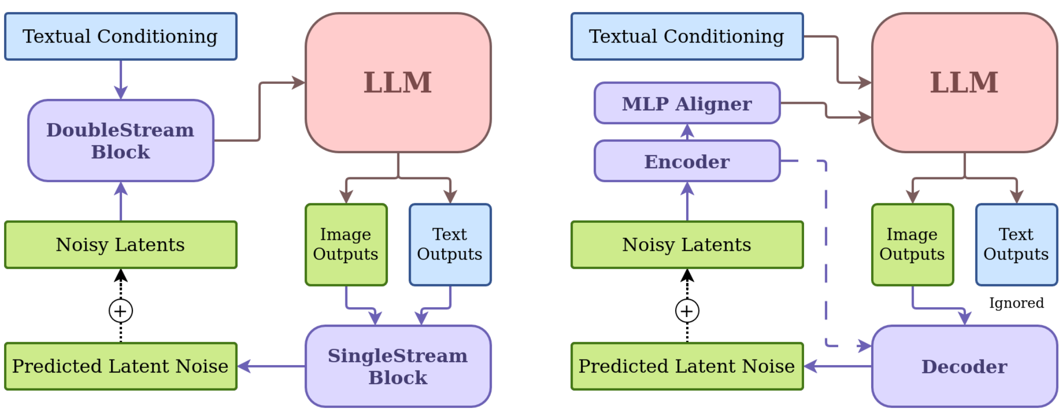

Figure 3: Comparison of our proposed (left) and the original Janus architectures (right). Our model employs a Dual-Stream Block Encoder for joint text and image encoding, while the original method utilizes an MLP Aligner and Encoder to map latents before passing them to an LLM. The single-stream decoder in our approach processes both noise and text LLM outputs, while the original decoder handles only noise.

A.2 Details

Dual-stream encoder (MM-DiT-style) We adopt a dual-parameterization transformer encoder in the spirit of MM-DiT: two weight sets (noise/image stream and text stream) with joint attention over the concatenated token sequence, and optional QK-normalization for stability Esser et al. [2024]. In practice, text and image/noise embeddings are concatenated and processed by per-stream blocks while sharing the attention field over the combined sequence, following SD3's MM-DiT design. A simple sinusoidal positional embedding is applied to the conditioning input before the encoder.

Widths. All blocks use the same hidden size as our prior LLM (TinyLlama) and multi-head attention with n_h=32.

Depth and blocks. d_e=3 identical blocks. Each block (per stream) applies: (i) RMSNorm; (ii) multi-head self-attention over the concatenated sequence with per-stream {Q,K,V,Out}; (iii) RMSNorm; (iv) MLP; (v) residual connections.

Timestamp token. Following JanusFlow, we append a *single learnable time token* representing the current timestamp t to the end of the sequence at generation time.

Parameter accounting. With the above widths, compact feed-forward choice, and dual streams, the encoder is $\approx 1.8 \times 10^8$ parameters in total (three dual-stream blocks), measured from our implementation.

Single-stream decoder (ViT-22B-style) We use a single-stream transformer decoder with *parallel* Attention+MLP (applied to the same pre-normalized input and summed), optional QK-normalization, and biases omitted in QKV/LN (kept in MLP), adapted for latent-space decoding. The final linear layer maps hidden states to $c \cdot p^2$ channels for unpatching to VAE latents (p=2, latent channels $c = 4$).

Depth and blocks. d_d=2 identical blocks with the same hidden width and number of heads setting as the encoder.

Parameter accounting. The decoder is $\approx 6.0 \times 10^7$ parameters in total (two single-stream blocks plus output head).

Total parameters Encoder ($\sim$ 180M) + decoder ($\sim$ 60M) + embeddings/patching/projections ($<$ 10M) $\Rightarrow$ total $\approx$ **240**M parameters.

B Training Details

Hardware & framework. 2 nodes $\times$ 8 NVIDIA H100 GPUs; PyTorch Distributed Data Parallel (DDP).
Runtime. End-to-end training $\sim$ 8.5 days.
Resolution. All training images are 384$\times$384.
Schedule & hyperparameters. Two-stage recipe; full settings in Table 1, overview in Fig. 4.

Hyperparameter	Stage 1	Stage 2
Learning Rate	1×10^{-4}	
Optimizer	AdamW ($\beta_1 = 0.9,\ \beta_2 = 0.999$)	
Scheduler	Constant	Constant
Training Steps	10k	100k
Warmup Steps	2000	
Effective Batch Size	512	
Weight decay	0.0	

Table 1: Hyperparameters used for Stage 1 and Stage 2 of training. Shared values are centered, while stage-specific values are listed separately.

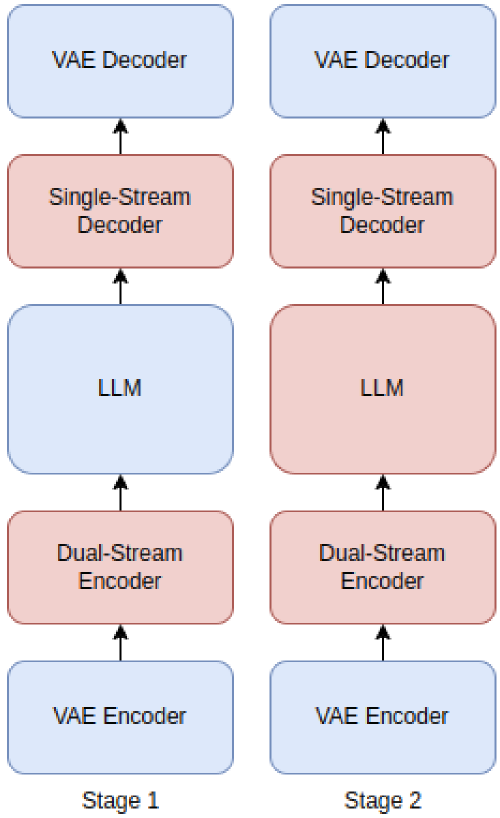

Figure 4: Stage 1 (left): Align the encoder-decoder with the pretrained LLM's space, without training the LLM. Stage 2 (right): Fine-tune the entire model, including the LLM, for image generation. (blue - weights are frozen, red - component is being trained)

B.1 U-ViT Baseline Implementation

For completeness, we include details of the U-ViT baseline used in Section 6. We adopt the `U-ViT-h1536-d42-n16` configuration from Bao et al. [2023], which corresponds to a model with hidden dimension 1536, depth 42, and 16 attention heads. All other architectural and training settings (data preprocessing, optimizer, learning rate schedule, and sampling protocol) are identical to those used in our main experiments.

The implementation is based directly on the official code release provided by the authors of U-ViT[1]. We scaled this configuration to approximately 1.3B parameters, consistent with the efficient scaling guidance of Li et al. [2024].

C Sampling Protocol

For all reported quantitative and qualitative experiments, we used the following sampling configuration unless otherwise noted:

- **Number of steps:** 30
- **Sampler:** Euler solver
- **Classifier-Free Guidance (CFG) scale:** 2.0

[1]`https://github.com/baofff/U-ViT`

C.1 Background on the Euler Sampler

Euler sampling is one of the simplest numerical solvers for ordinary differential equations (ODEs), commonly used in rectified flow and diffusion-based generative models. It approximates the continuous trajectory of the rectified flow by discretizing time into steps and updating the latent representation using local gradient information. While simple, Euler's method is computationally efficient and provides stable results when combined with rectified flow guidance.

C.2 Background on Classifier-Free Guidance (CFG)

Classifier-Free Guidance (CFG) is a technique that improves conditional generation by interpolating between unconditional and conditional model outputs. Specifically, the model is run once with the conditioning signal (e.g., text prompt) and once without it; the two predictions are linearly combined using a guidance scale γ. A higher γ encourages stronger alignment with the condition at the cost of sample diversity. In our experiments, $\gamma = 2.0$ provided a good balance between faithfulness to text and image quality.

CFG sweep. We conducted a sweep over classifier-free guidance (CFG) strengths $\{1.5, 2.0, 2.5, 3.0\}$ using our default Euler sampler with 30 steps on MJHQ-FID-30k. FID and CLIP scores (mean of three runs) showed that **CFG = 2.0** consistently offered the best trade-off between fidelity and semantic alignment. At lower CFG (1.5), prompt adherence weakened; at higher values (2.5, 3.0), fidelity degraded with oversaturation artifacts. Therefore, all main experiments in the paper report results at CFG = 2.0.

C.3 Algorithm: Sampling Procedure

Algorithm 1 Rectified Flow Sampling with Euler and CFG

Require: Initial noise latent x_0, text embedding c, steps $T = 30$, CFG scale $\gamma = 2.0$

for $t = 0$ to $T - 1$ **do**
  Compute conditional velocity $v_{\text{cond}} = f(x_t, c, t)$
  Compute unconditional velocity $v_{\text{uncond}} = f(x_t, \varnothing, t)$
  Combine: $v = v_{\text{uncond}} + \gamma \cdot (v_{\text{cond}} - v_{\text{uncond}})$
  Euler update: $x_{t+1} = x_t + \Delta t \cdot v$
end for
Return x_T (decode with VAE)

D Qualitative results

Figure 5: Sample images generated by our model. These images demonstrate the enhanced image generation capabilities achieved using our LLM-backed Rectified Flow with transformer-based encoder and decoder.

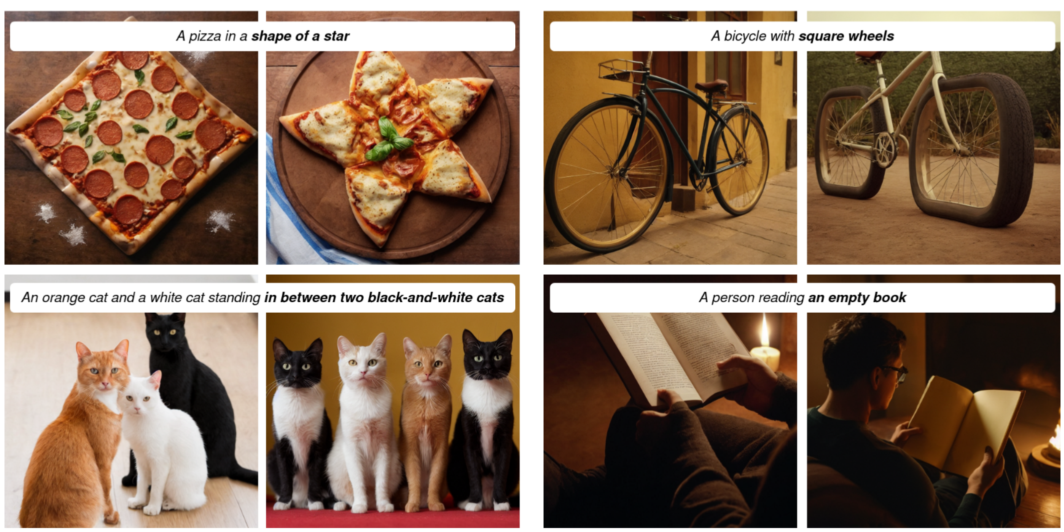

Figure 6: Qualitative comparison of generated images. Each pair shows an image generated by the baseline (left) and our improved model (right) for the same input prompt. Our model demonstrates enhanced spatial coherence, finer detail preservation, and improved text–image alignment.

E Human Study Design and Details

We conducted a human evaluation study to assess prompt alignment and perceptual quality of images generated by our model compared to baselines.

E.1 Participants

The study involved 52 participants, all of whom were employees of our company. Participation was strictly voluntary, and no monetary or material compensation was provided. Distributed using corporate email.

E.2 Procedure

Each participant was shown a set of 40 randomly selected prompts with two corresponding images (ours vs. baseline). The order of images (left/right) was randomized per comparison to mitigate positional bias. Similarly, the order of prompts was randomized for each participant to reduce ordering effects. Participants were not informed about which model generated each image, ensuring that comparisons were conducted in a blind setting.

E.3 Instructions

For each prompt and pair of images, participants were asked to answer two multiple-choice questions:

1. **Prompt alignment:** "Which image looks more representative of the text shown above and faithfully follows it? (Left / Right)"
2. **Aesthetic quality:** "Given the prompt, which image is of higher quality and aesthetically more pleasing? (Left / Right)"

Responses were analyzed using a two-sided binomial test with null hypothesis $p = 0.5$, and 95% Wilson confidence intervals were computed for the preference rate of our model.

E.4 Question 1: Prompt Faithfulness

> *"Which image looks more representative of the text shown above and faithfully follows it? (Left / Right)"*

- Votes for our model: 1227
- Votes for baseline: 853
- Win-rate for our model: 59%
- Binomial test ($p = 0.5$): p-value $= 1.198 \times 10^{-16}$
- Wilson confidence interval (95%): $(0.5686, 0.6109)$

E.5 Question 2: Aesthetic Quality

> *"Given the prompt, which image is of higher quality and aesthetically more pleasing? (Left / Right)"*

- Votes for our model: 1082
- Votes for baseline: 998
- Win-rate for our model: 52%
- Binomial test ($p = 0.5$): p-value $= 0.0344$
- Wilson confidence interval (95%): $(0.4987, 0.5416)$

E.6 Ethics and Consent

All participants gave informed consent to voluntarily take part in the study. The study was conducted internally and did not involve external subjects or sensitive data. No personally identifiable information was collected, and results were aggregated anonymously. Since the evaluation was conducted on a voluntary basis with company employees and did not involve vulnerable populations or sensitive topics, no formal IRB approval was required.

E.7 Analysis

For each question, we aggregated the votes across participants and computed the percentage of preferences for our model versus the baseline. Statistical significance was assessed using a binomial test at the $p < 0.05$ level.

F Dataset

F.1 Sources, composition, and preprocessing

Sources & licenses. DALL-E 3 High-Quality Captions (MIT) and a "laion-coco-aesthetic" subset Liu derived from LAION-5B (Apache-2.0) Schuhmann et al. [2022].
Composition. We mix datasets at a 1:2 ratio (DALL-E:LAION–COCO) to form **3M** image–text pairs. Typical source resolutions are 1024^2 and 1792×1024.
Preprocessing. Center-crop and resize all images to **384×384**; captions are used as provided by the datasets.

F.2 Additional characteristics of the sources

DALL-E 3 High-Quality Captions. This corpus consists primarily of AI-generated images (largely DALL-E 3, with contributions from MidJourney and Stable Diffusion) paired with detailed, model-written captions. It spans a wide range of subject matter (objects, scenes, portraits), artistic styles (photorealism, digital art, watercolor, anime), and compositional patterns (single-object, multi-entity, relational prompts). The release is filtered for duplicates, non-AI content, and inappropriate material. The captions are typically longer and more attribute-rich than web captions, which we find beneficial for conditioning.

LAION–COCO aesthetic subset. We draw from the "laion-coco-aesthetic" split Liu, a high-aesthetic slice of LAION-5B Schuhmann et al. [2022] scored via CLIP-based aesthetics Schuhmann. Images are real-world and higher-variance in camera viewpoint, lighting, and texture statistics compared to synthetic sources. Captions are synthetic but concise. We select a higher-aesthetic tranche to complement DALL-E imagery, balancing stylized variety with natural image statistics.

F.3 Benchmark prompt isolation.

All evaluation prompts were drawn from the MJHQ FID-30k benchmark. Prior to training, we hashed all evaluation prompts and removed any entries from our training corpus that contained exact or near-duplicate text spans (using case-insensitive matching and cosine similarity thresholding at 0.95 over sentence embeddings).

F.4 Prompt Leakage Prevention

A critical concern in evaluating text-to-image generative models is the potential for *prompt leakage*, where benchmark prompts inadvertently overlap with training data. Such overlap may lead to inflated performance metrics and undermine fair comparison.

To mitigate this risk, we adopted the following protocol:

1. **Dataset provenance and licensing.** Our training data consisted of the DALL-E 3 High-Quality Captions dataset (MIT license) and a curated subset of the LAION-COCO dataset (Apache 2.0 license).
2. **Randomized verification.** A random sample of 1,000 training prompts was manually inspected against the evaluation set to confirm absence of near-overlap in phrasing and semantic structure.

F.5 Human Study Prompts

For the human preference study, evaluation prompts were not taken from any existing dataset but were instead **generated with GPT-4o**. This approach allowed us to create diverse and natural instructions while avoiding contamination from commonly used benchmarks.

To ensure reproducibility and fairness:

1. **Prompt generation.** We instructed GPT-4o with prompt designed to elicit varied, semantically rich descriptions spanning objects, styles, and compositional requirements:

   "Generate 40 diverse text prompts suitable for evaluating text-to-image models. Prompts should vary in style (photorealistic, artistic, abstract), complexity (single object vs. multiple entities with relations), and domains (nature, urban, science fiction, cultural). Avoid clichés and keep prompts between 10–25 words."

2. **Leakage avoidance.** After prompt generation, we ran the same hashing and semantic similarity filtering procedure as with the benchmark set to confirm that none of these GPT-4o-generated prompts overlapped with our training data corpus.

G Key Evaluation Metrics

G.1 Fréchet Inception Distance (FID) Computation

For distribution-level image quality, we report the Fréchet Inception Distance (FID). Specifically:

- **Backbone:** Inception-V3, pool3 activations (2048-D).
- **Computation:** Let μ_r, Σ_r be the mean and covariance of reference features, and μ_g, Σ_g those of generated features. FID is

$$\mathrm{FID}(\mathcal{R}, \mathcal{G}) \;=\; \|\mu_r - \mu_g\|_2^2 \;+\; \mathrm{Tr}\big(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}\big).$$

  For numerical stability we add a small diagonal term ($10^{-6} I$) to covariances prior to the matrix square root.

G.2 CLIP Score Computation

For image–text alignment evaluation, we report the CLIP score as a cosine similarity between image and text embeddings, following standard practice. Specifically:

- **Model:** We use the OpenAI CLIP ViT-L/14@336px encoder.
- **Computation:** Given a generated image I and its corresponding text prompt T, we obtain the normalized image embedding $f(I)$ and text embedding $g(T)$. The CLIP score is then

$$\mathrm{CLIP}(I, T) = \frac{f(I) \cdot g(T)}{\|f(I)\| \, \|g(T)\|} \in [0, 1].$$

  Note: All CLIP scores reported in this paper are multiplied by 100 for readability.
- **Aggregation:** For each model, we report the mean CLIP score over the entire evaluation set (30k prompts from MJHQ).
- **Variance reporting:** We additionally compute the standard deviation across random seeds.

H Representational Analysis: Protocols and Additional Results

H.1 Common protocol

Linear CKA. Let $X, Y \in \mathbb{R}^{N_{\text{tok}} \times d}$ be token-stacked representations (same indices) after centering over tokens. We use linear CKA

$$\mathrm{CKA}(X, Y) \;=\; \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \;\cdot\; \|Y^\top Y\|_F}.$$

We *global-average-pool* token features before CKA. For prompt p and timestep t_k, compute $\mathrm{CKA}_p(t_k)$. For seed s, $\overline{\mathrm{CKA}}_s(t_k) = \frac{1}{N}\sum_{p=1}^{N} \mathrm{CKA}_p(t_k)$; then average across seeds $\overline{\mathrm{CKA}}(t_k) = \frac{1}{S}\sum_{s=1}^{S} \overline{\mathrm{CKA}}_s(t_k)$ and report mean$\pm$1 SE over S=5 seeds.

Data, sampler, seeds. MJHQ-30k, random 2k-prompt subset; Euler solver, 30 steps; CFG $= 2.0$; S=5 seeds.

Controls. To preclude trivial collapse, we monitor per-feature variance of $E^{\text{noise}}(t_k)$ over prompts and confirm it remains within $\pm 10\%$ of the ConvNeXt condition for all k. Shuffling captions across images substantially reduces CKA (plots omitted), indicating condition sensitivity.

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